

Green Lasers Overcome Last Barrier to Brilliant Displays

Up to now only red and blue laser diodes were commercially available, while green lasers were obtained from infrared lasers by a two-step complicated and costly process. Lasers that generate green light directly plug the green gap in the red-green-blue triad needed for full-colour laser projection and display



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Solidstate lasers can produce light in red and blue parts of the visible spectrum, generating laser light in all colours except green. But recent research work suggests that this 'green gap' could be plugged. New techniques for growing laser diodes could soon make brilliant full-spectrum displays a reality.

Plugging the green gap in the red-green-blue triad needed for full-colour laser projection and display would help speed the introduction of laser projectors for televisions and movie

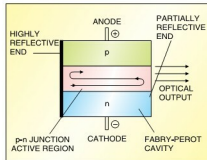


Fig. 1: Laser diode

theatres, which will display much richer colours than other systems, and tiny handheld projectors as in cellphones. High-power green diodes might be

employed in such applications as DNA sequencing, industrial process control and underwater communications.

The familiar green laser pointers used by lecturers employ a complicated two-step process to generate light. Semiconductor lasers inside these devices emit infrared radiation having a wavelength of around 1060 nanometres. This radiation then pumps a crystal that oscillates at half this wavelength – 530 nanometres, which corresponds to green part of the spectrum. The process is costly, inefficient and imprecise; the second crystal can heat up, altering the wavelength of the resultant green light. Lasers that generate green light directly would avoid this problem.

Multiquantum-well fundamentals

Fig. 1 shows the scheme of a homo-junction laser where a single p-n junction is formed from only a single-crystal semiconductor material. There is a highly reflective surface at one end and a partially reflective surface at the other end. The p-n junction is forward-biased by an external voltage source. As electrons move through the junction, recombination occurs just as in an ordinary LED light emitter.

To achieve laser action, it is necessary to contain photons within the laser medium and maintain the conditions for coherence. The optical cavity, as shown in Fig. 1, is known as Fabry-